



Predicting Seizures

Researchers develop a technique to predict epileptic seizure periods; similar to weather forecasts. <http://digit.in/jan21-23>



Ice Age Puppy

57,000 year-old wolf puppy specimen found frozen in Yukon permafrost. <http://digit.in/jan21-24>

individual cells in synchronisation or independently, by controlling the magnetic field on each of the plates. These compressing and expanding cylinders, with individual panels that can be controlled can further be fashioned into legs, or passive sensors for sensing the environment. For example, on meeting a wall, a probe made up of the Kresling pattern would collapse, providing an input to the robot to make the next move. The research pushes the boundaries of what soft robotics can achieve.

LASER CONTROLLED ROBOTS

It looks like the bloodstreams of future humans are going to be accentuated in all sorts of ways. Apart from the programmable drugs, researchers are also exploring the possibility of introducing tiny bacteria-sized microbots, which can themselves be loaded with sensors and drug delivery platforms in the future. At the present stage of the research, the focus is on miniaturisation, and creating cell sized robots that can work with conventional power sources, such as electricity. Researchers from Penn Engineering and Cornell have managed to do just that. The robots are roughly the size of paramecium. The robots are made with standard semiconductor processing, and can be made to walk using conventional electronic signals. The circuits for the robot are etched into silicon wafers, similar to the way conventional computer chips are manufactured.

The legs are added by atomic deposition, and are extremely thin strips of platinum, only a few dozen atoms in thickness. The platinum curls up when actuated, so rigid panels are added to give more form to the legs, and determine how exactly the legs bend when actuated.

More advanced versions of these cell sized robots can one day be used for various biomedical applications within the human body. The robots, just like protozoa, is invisible to the naked eye, and measures about 40 microns wide, 40-70 microns long and about 5 microns thick. The design of the robot is relatively simple, there are four photovoltaic cells in the torso, that can use laser signals to control the tiny actuators for each of the legs. The robots can move at about 30 microns per second, which is still too slow to be put to any good use. However, future improvements in the design, especially the bioinspired kind is expected to make the robot much faster.

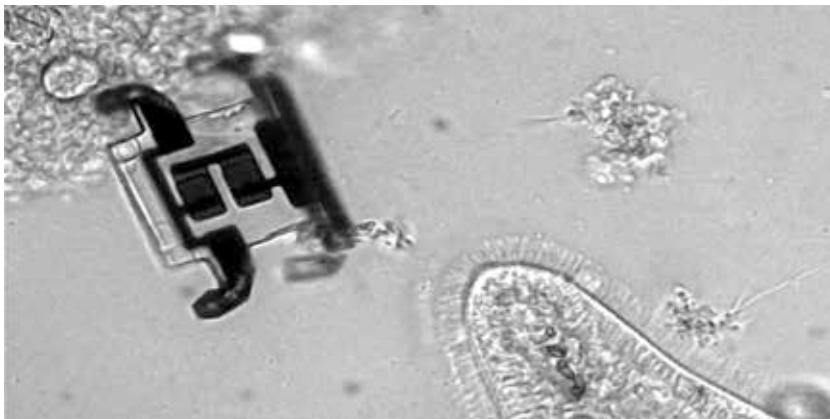
BACKFLIPPING THROUGH TERRAINS

If you remember the movement of TARS from Interstellar, the motion is suitable for navigating a wide range of difficult terrains. Researchers have implemented the same basic idea on a much larger scale, where the terrain is the internal surfaces of organs in living human beings. Researchers from Purdue have demonstrated the capability of a tiny rectangular robot that tumbles through the colons of



The microbot on a US penny

mice and pigs. Considering all the matter in the colons and the large number of different moving objects, it is a tough terrain for a robot to navigate, but it manages because of a clever trick. The robot is actuated externally by magnetic fields. David Cappelleri, one of the developers of the robot, explains, “when we apply a rotating external magnetic field to these robots, they rotate just like a car tire would go over rough terrain. The magnetic field also safely penetrates different types of mediums, which is important for using these robots in the human body.” The robots are made up of polymer and metal, and are biocompatible, which means that they will not damage the delicate tissue within the body, or cause any adverse reactions. Conventional machinery can be used to produce large swarms of these tumbling robots. Researchers intend to use the microbots as drug delivery platforms, but it is also possible to make them performing colonoscopies by collecting tissue samples. Once it has navigated to a target location, the robot can stay in place for weeks, delivering drugs to the site at periodic intervals. 



One of the cell sized laser controlled robots